

IKB42603 – CLOUD COMPUTING SECURITY ESSENTIALS

Lab 0: Environment Setup Report

Course Information

Course Name: IKB42603 – Cloud Computing Security Essentials

Lab: Lab 0 – Environment Setup

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Objective

The objective of this lab is to prepare the local environment required before performing the Cloud Computing Security Essentials laboratory exercises. The setup includes installing, verifying, and configuring Docker, AWS CLI v2, kind, kubectl, OpenSSL, LocalStack, and a local Kubernetes cluster. The environment is configured to run locally using LocalStack as the AWS simulator and kind for Kubernetes.

Step 1: Install and Verify Docker

Docker was verified to ensure that the container runtime was available for running LocalStack and Kubernetes services.

Actions Performed

- Verified Docker installation.
- Executed the Hello World container to confirm Docker was functioning correctly.

Verification Commands

```
docker --version  
docker run --rm hello-world
```

```
user@LAPTOP-SN02HB52 MINGW64 ~  
$ docker --version  
Docker version 29.6.2, build dfc4efb  
  
user@LAPTOP-SN02HB52 MINGW64 ~  
$ docker run --rm hello-world  
Unable to find image 'hello-world:latest' locally  
latest: Pulling from library/hello-world  
4f55086f7dd0: Pull complete  
d5e71e642bf5: Download complete  
Digest: sha256:7f4da0fc94bcece205a8c0b6f4d11c8196924654ffe5c4d1aa439b7f632048b2  
Status: Downloaded newer image for hello-world:latest  
  
Hello from Docker!  
This message shows that your installation appears to be working correctly.
```

Result

Docker was successfully verified, and the Hello World container executed successfully.

Step 2: Install and Verify AWS CLI v2

AWS CLI v2 was verified to enable communication with LocalStack using AWS commands.

Actions Performed

- Verified AWS CLI installation.
- Confirmed the installed version.

Verification Commands

```
aws --version
```

```
MINGW64:/c/Users/user  
  
user@LAPTOP-SN02HB52 MINGW64 ~  
$ aws --version  
aws-cli/2.36.16 Python/3.14.6 windows/11 exe/AMD64  
  
user@LAPTOP-SN02HB52 MINGW64 ~  
$
```

Result

AWS CLI v2 was successfully verified.

Step 3: Install and Verify kind and kubectl

kind and kubectl were verified to support the creation and management of a local Kubernetes cluster.

Actions Performed

- Verified kind installation.
- Verified kubectl installation.

Verification Commands

kind --version

kubectl version --client

```
user@LAPTOP-SN02HB52 MINGW64 ~  
$ kubectl version --client  
Client Version: v1.34.1  
Kustomize Version: v5.7.1  
  
user@LAPTOP-SN02HB52 MINGW64 ~  
$
```

```
user@LAPTOP-SN02HB52 MINGW64 ~  
$ kind --version  
kind version 0.32.0  
  
user@LAPTOP-SN02HB52 MINGW64 ~  
$
```

Result

Both kind and kubectl were successfully verified and are ready for Kubernetes operations.

Step 4: Install and Verify OpenSSL

OpenSSL was verified to ensure cryptographic tools were available for future laboratory exercises.

Actions Performed

- Verified OpenSSL installation.

Verification Commands

openssl version.

```
user@LAPTOP-SN02HB52 MINGW64 ~  
$ openssl version  
OpenSSL 3.5.4 30 Sep 2025 (Library: OpenSSL 3.5.4 30 Sep 2025)  
  
user@LAPTOP-SN02HB52 MINGW64 ~  
$
```

Result

OpenSSL was successfully verified.

Step 5: Start and Verify LocalStack and Verify LocalStack Health

LocalStack was started to emulate AWS cloud services in a local environment.

Actions Performed

- Started the LocalStack container.
- Verified that the LocalStack container was running successfully.

Verification Commands

```
docker run -d --name localstack -p 4566:4566 localstack/localstack:3.8.1
```

```
docker ps
```

```
curl http://localhost:4566/_localstack/health
```



```
user@LAPTOP-SNO2HB52 MINGW64 ~
$ docker ps
CONTAINER ID   IMAGE                                PORTS                                COMMAND                                CREATED
STATUS
af00368dd5aa   localstack/localstack:3.8.1         "docker-entrypoint.sh"              29 seconds ago
Up 28 seconds (healthy)    0.0.0.0:4566->4566/tcp, [::]:4566->4566/tcp
localstack
```

Result

LocalStack was successfully started, and the container status was healthy.

Step 6: Create and Verify Kubernetes Cluster

A local Kubernetes cluster was created using kind and verified using kubectl.

Actions Performed

- Created the Kubernetes cluster.
- Verified cluster information.
- Verified node status.

Verification Commands

```
kind create cluster --name ccse
```

```
kubectl cluster-info --context kind-ccse
```

```
kubectl get nodes
```

```
user@LAPTOP-SN02HB52 MINGW64 ~  
$ kubectl get nodes  
NAME                STATUS    ROLES    AGE   VERSION  
ccse-control-plane  Ready    control-plane  70s   v1.36.1
```

Result

The Kubernetes cluster was successfully created, and the control-plane node was in the Ready state.

Step 7: Configure AWS CLI for LocalStack

The AWS CLI was configured with test credentials and connected to the LocalStack endpoint.

Actions Performed

- Configured AWS CLI.
- Set the LocalStack endpoint.
- Verified the connection using the STS service.

Verification Commands

```
aws configure set aws_access_key_id test
```

```
aws configure set aws_secret_access_key test
```

```
aws configure set region us-east-1
```

```
EP='--endpoint-url=http://localhost:4566'
```

```
aws $EP sts get-caller-identity
```

```
user@LAPTOP-SN02HB52 MINGW64 ~  
$ aws $EP sts get-caller-identity  
{  
  "UserId": "AKIAIOSFODNN7EXAMPLE",  
  "Account": "000000000000",  
  "Arn": "arn:aws:iam::000000000000:root"  
}
```

Result

The AWS CLI was successfully configured and connected to the LocalStack environment.

Conclusion

The Lab 0 environment setup was completed successfully. Docker, AWS CLI, kind, kubectl, OpenSSL, LocalStack, and the Kubernetes cluster were successfully verified and configured. The local cloud environment is now ready for the remaining Cloud Computing Security Essentials laboratory exercises.

Summary

Component	Verified Version /Status	Purpose	Status
Operating System	Windows 11	Host system for all development tools	Configured
Docker	Docker 29.6.2	Container runtime for LocalStack and Kubernetes	Verified
AWS CLI	AWS CLI 2.36.16	Command-line interface for AWS and LocalStack	Verified
kind	kind 0.32.0	Create and manage local Kubernetes clusters	Verified
kubectl	Client Version 1.34.1	Kubernetes cluster management	Verified
OpenSSL	OpenSSL 3.5.4	Cryptographic utilities	Verified
LocalStack	Community Edition v3.8.1 (Healthy)	Local AWS cloud service emulator	Running

Kubernetes Cluster	<code>ccse-control-plane</code> (Ready)	Local Kubernetes environment	Running
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